The missing value plot

How many features are missing in no sample, in one sample, in two, and so on to the end of the experiment.

AT A GLANCE

Where	Completeness → Missing Value Plot
Data levels	Protein, phosphosite, phosphoprotein, peptide, precursor
Needs	An uploaded dataset and its annotation. Peptide and precursor additionally need a chosen abundance column.
Data source	The matrix as uploaded. Completeness has no Filtered or Imputed choice: it describes what arrived, before any decision taken about it.
One bar	The number of features missing in exactly that many samples.
Table	No. The same counts, with percentages, are on the Tables page.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The bar over 0 counts features quantified in every included sample. Bars to its right count increasingly incomplete features. Completeness alone does not decide whether to filter or impute: the experimental design, location of the gaps and plausible reasons for missingness also matter.

The plot says how *much* is missing per feature, not *where*. A feature with one missing value is one bar to the right whether the gap is in the first sample or the last; the heatmap on the

next page is what distinguishes them.

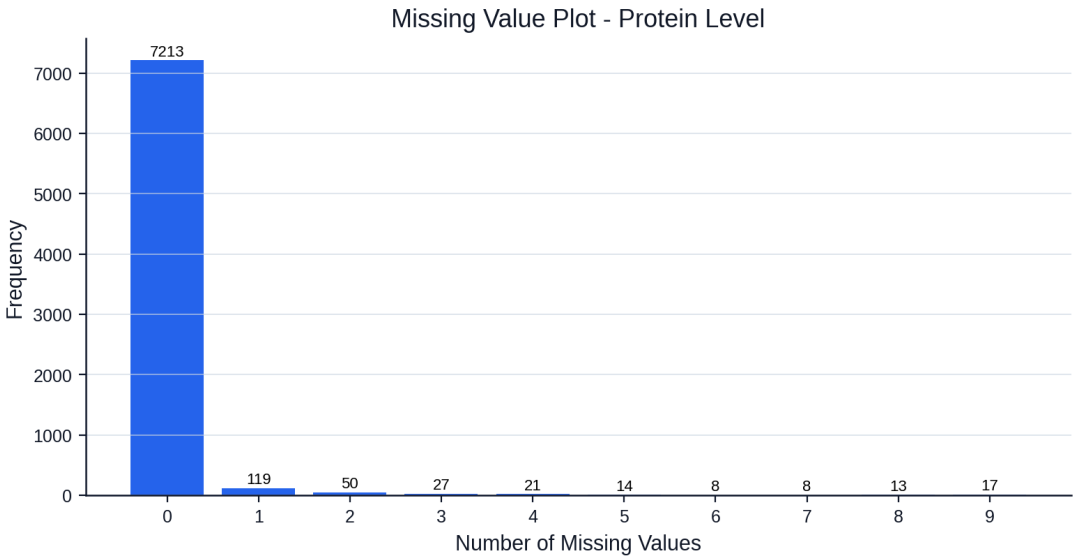


Figure 1.1. Protein level on the example project, ten samples: 7213 proteins are missing in none of them, and the tail from 1 to 9 holds the 277 that are missing somewhere. The axis stops at 9, one short of the sample count.

Reading it

The bullets below read a figure from a larger experiment — 96 samples rather than the ten above — so that the tail has something in it to read.

- 3663 proteins sit over 0: they carry a value in all 96 samples.
- 448 proteins sit over 1: 95 samples measured them and one did not. Which one is not shown here.
- 27 proteins sit over 95: each was reported in a single sample.
- There is no bar over 96, and there cannot be one. A feature missing everywhere is dropped before the counting, so the axis stops one short of the sample count.

> **EASY TO GET WRONG**
>
> The x axis is a list of the counts that actually occur, not a number line. If no feature is missing in exactly 2 samples, the axis runs 0, 1, 3 with no gap where 2 would be, and the bars either side of the absent value sit next to each other. Read the tick labels before reading the shape, and read the Tables page when the exact numbers matter.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to describe: Protein, Phospho, Phosphoprotein, Peptide or Precursor. Only the levels actually uploaded are offered; Phosphoprotein appears once a collapse has been run, and Peptide and Precursor only on this page, from the full report.	Protein	page
Toggle Text	Prints the count over each bar.	on	page
Bin missing values	Folds the long tail into one bar. With the value *b* above zero, every feature missing in more than *b* samples is counted into a single >b bar and the axis runs 0 to b and then >b. Zero means no binning.	0	page
Abundance Column	Peptide and precursor only. Which quantity column of the report is pivoted before the missing values are counted. Precursor.Quantity is chosen automatically when the report has it; otherwise, with more than one candidate column, no plot is drawn until one is picked.	the report's own	page
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, the resolution between 72 and 600.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size, weight, colour and position. The title is written from the data level: Missing Value Plot - Protein Level and its equivalents for phosphosite, phosphoprotein, peptide and precursor.	shown	appearance
X axis, Y axis	The text and styling of Number of Missing Values and Frequency, and of the tick labels under the bars.	plot's own	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Frame and grid	Which of the four spines are drawn, and the grid behind the bars. This plot opens with all four spines and a horizontal dashed grid.	four spines; horizontal dashed grid, #b0b0b0, width 0.8	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency.	Default, white, opaque	appearance

NOTE

Bar Color and **Text Size**, which the earlier manual listed beside the plot, are gone from the page. The renderer still reads a bar colour and a label size, but neither the **Bars** card nor the **Value labels** card is offered on this page, so the bars are drawn in the default blue and the printed counts at size 8. The **Bars** and **Value labels** cards are offered here: this renderer reads both the bar colour and the label size.

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

HOW IT IS COMPUTED

The matrix is restricted to the sample columns named in the annotation. A value is missing when it is NA, when it is not a number, or when it is infinite; a zero counts as missing too, because a zero in a linear intensity means not detected. The one exception is a dataset uploaded already on the $\log_2$ scale, where zero is a real measurement — an intensity of 1 — and is kept.

For feature i over n sample columns,

$$m_i \;=\; \sum_{j=1}^{n} \mathbf{1}\big[x_{ij} = \mathrm{NA}\big]\,, \qquad h_k \;=\; \#\{\, i \,:\, m_i = k \,\}.$$

Features with $m_i = n$ are removed before h_k is formed, so k runs from 0 to $n - 1$ and the height of a bar is the number of features missing in exactly that many samples. Only the values of k that occur are drawn.

WITH BINNING

With **Bin missing values** set to $b > 0$, every m_i above b is relabelled $>\mathrm{b}$ before counting, and the axis is fixed to $0, \ldots, \mathrm{b}, >\mathrm{b}$. The exclusion of the missing-everywhere features happens first, on the numbers, so the $>\mathrm{b}$ bar never quietly absorbs them.

PEPTIDE AND PRECURSOR

The full report is long, so it is pivoted first: rows are the stripped sequence at peptide level and the precursor identifier at precursor level, columns are the run file names mapped onto the annotated samples, and the values are the chosen abundance column. Duplicate entries for the same feature and run are reduced with the maximum — not the first row, which would depend on the order the file happened to be written in. Zeros then become NA, and the counting is the same.

The example worked

Three proteins over three samples:

THE MATRIX, AND THE TWO BARS IT GIVES

	A1	A2	B1	m_i
Protein 1	3.34	3.54	NA	1
Protein 2	3.34	3.54	2.67	0
Protein 3	3.34	NA	2.67	1

Number of missing values	Frequency
0	1
1	2

Two bars, not three. There is no bar over 2 because no protein is missing in exactly two samples, and there could be none over 3 in any case: that is the sample count. The Tables page prints the same two rows with their percentages, 33.33 % and 66.67 %.

Edge cases

- A feature missing in every sample never appears. It occupies a row of the matrix, but it is left out of this plot, the heatmap and the Tables page alike, and the Tables page says how many such features were excluded.
- A missing-value count that no feature has gets no bar and no tick.
- A dataset with nothing missing draws a single bar over 0.
- At peptide and precursor level, nothing is drawn until an abundance column is chosen, and the plot then describes that column alone: a second abundance column in the same report can give a different figure.
- Binning changes the picture but not the underlying counts: the same dataset summed over the bars gives the same total at every setting of *b*.

Interpreting the pattern

A tall bar over 0 means many features were observed in every selected sample. A rise near the other end means many were observed only sparsely. Neither shape is universally right or wrong. A peak around half the sample count could come from features restricted to one of two equal-sized conditions, or from missing values scattered across both. This histogram cannot distinguish them; the missing-value heatmap supplies the sample and condition pattern.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = missing_value_plot(data, meta, data_scale="log2")
```

R

```
fig <- missing_value_plot(data, meta, data_scale = "log2")
```

There is no data level to choose: the level is the frame handed in, so the phosphosite and phosphoprotein figures are the same call on those frames, and the title is written from the frame's own identifier column. data_scale carries the convention the page applies silently — "log2" keeps a zero as a measurement, "linear" reads it as a non-detection — and bin is the binning control, at 0 unless it is given.

5 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen: the displayed plot is fitted to the window, the downloaded one to the numbers. Choose PDF for a figure that will be scaled or edited, PNG for a slide. There is no table under this plot; the same counts, with percentages, download as CSV from the Tables page.

Behind this page

Function	What it produces
missing_value_plot	The plot for wide-format data: protein, phosphosite and phosphoprotein
missing_value_plot_pep	The peptide-level version, from the long-format report
missing_value_plot_prec	The precursor-level version, from the long-format report